

Documentation Leibinger-Jet3 Script language

V1.6.0.15

Summary

Language layout	3
Syntax:.....	3
Coordinate system	3
Keywords	4
Beginning of Script	5
End of Script.....	5
Layout of parameter lists	6
JLPAR parameter list (Job list parameter)	6
Print Height	6
PrintGo Signal	6
PrintGo-Gate Signal	6
Encoder signals	7
Rotation, mirror-inverted	7
Encoder resolution.....	7
Shifting Change of date	7
Backward Lock	7
BEGINJOB parameter list:.....	8
ID-number:	8
ENDJOB parameter list.....	8
JOBPAR parameter list (Job parameter)	9
PrintGo-delay	9
PG-repetition	9
PG-distance	9
Stroke distance	9
Print Mode.....	10
Flight time compensation	10
Stroke monitoring.....	11
PrintGo monitoring	11
PGHoldOff	11
IdentNr CameraJob	11
CameraJob Name	11
Max. products between LJ3 and camera	11
Product offset between LJ3 and camera.....	11
Cameratrigger-delay	12
Print stop interrupts print-out immediately	12
Indexcounter of previous camera job data	12
CNT parameter list (Counter parameter)	13
Start value:.....	13
Initialisation value:	13
End value:.....	13
Increment:.....	13
Repetitions:.....	13
Counting event:	13

Reset	14
LoopMode	14
Repetitor	14
Don't print	14
Global counter	14
COD Parameter list (Barcode parameter)	15
Barcode type:	15
Position of clear figure	16
RPLFIG Parameter list	17
RPLDAY Parameter list	18
RPLMON Parameter list	18
RPLYEAR Parameter list	20
SHIFTS Parameter list until LJScriptversion V1.6.0.0	20
SHIFTS Parameterliste from LJScriptversion V1.6.0.1	21
EXTTXT Parameter list	21
TIME Parameter list	22
Layout Format	22
OBJ Parameter list (Object parameter)	24
ID-number:	24
Position:	24
Ltype:	24
Fonts:	25
Printing text	27
Character gap	27
Rotation, Mirror-inverted	27
Proportional:	27
Bold text:	27
Contrast print:	27
Inverse:	27
Alternate:	27
Prompt-Field:	27
PGJOB Parameter list (Batchjob)	28
ID-number:	28
Number of repeats:	28
EXTSEL Parameter list (Extern Job Select)	28
External applied number:	28
JOBORG Parameter list (Job organizer)	29
ID-number:	29
Distance:	29
EXAMPLE for a Script:	30
History	31

Language layout

The Script language LJScrip is a language to describe complete print layouts for the Leibinger-Jet 3.

LJScrip uses only ASCII-characters and as control characters only Carriage Return (CR) and Line Feed (LF).

Used are commands which have fixed relocated keywords. A parameter list follows. Further more the following comments can be inserted.

Syntax:

% comment

KEYWORD [Parameter list]

KEYWORD [Parameter list] % comment

KEYWORD [Parameter list]

Keywords are always written in capital letters.

The parameter list is specified in square brackets: [Parameter list]

Only one keyword is permitted per line.

The several parameters in the parameter list are separated by one ore more blanks, tabulators (Tab, Hex 09) or Carriage Return (0x0D) and Line Feed (0x0A).

Lists and texts which occur in the parameter list are specified in parenthesis ().

A backslash \ is prepend of the following characters if they should be part of the text:

) < \ % [] results in: \) \< \\ \% \[\]

Every element which is declarated refers to the last defined object.

Comments can be used. They always start with the percentage symbol % and end always at the end of line (CR and LF). They cannot be in the comment.

If the parameters have a limit value MAXINT, the values margin of 32 Bit Integer is valid. It goes from -2.147.483.648 up to +2.147.483.647

Floating point numbers are indicated with the decimal point. Power specifications do not exist.

Time is indicated in hours (hh) and minutes (mm) and/or seconds (ss). Separators are indicated in colons. Example: l: hh:mm:ss

Coordinate system

X is the transport direction. Units are always strokes

The 1. stroke has the X-coordinate X.0

Y is the direction in which the drops are deflected. Units are ink dots

The 1. ink dot is placed at the gutter tube and has the Y-coordinate Y.0

Keywords

Keywords	Meaning
BEGINLJSCRIPT	Indicates the beginning of a Script (job list)
JLPAR	Declares the job list parameter
BEGINJOB	Indicates the beginning of a job
JOBPAR	Declares the job parameter
OBJ	Declares object parameter
CNT	Declares counter objects
COD	Declares barcode objects
TIME	Declares date-/time objects
SHIFTS	Declares shift time objects
RPLDAY	Declares replacements for weekdays of date objects
RPLMON	Declares replacements for months of date objects
RPLYEAR	Declares replacements for years of date objects
ENDJOB	Indicates the end of a job
PGJOB	Declares Batch Job List
JOBORG	Declares job organizer list
EXTSEL	Declares external job select list
ENDLJSCRIPT	Indicates the end of a Script

Table 1: Keywords

Beginning of Script

Syntax: BEGINLJSCRIPT [Parameter list]

Every Script has to start with this keyword.

The parameter list contains only one text which has to have the version of the following format:

(Va.b.c.d)

Meaning:

a=Generation

b=Upgradings /Changes

c=Bugfix-counter

d=Index for special functions

Further parameters can follow but they can only be used as comments.

End of Script

Syntax: ENDLJSCRIPT [Parameter list]

Every Script has to end with this keyword.

The parameter list is not analyzed, can be also empty.

Layout of parameter lists

JLPAR parameter list (Job list parameter)

Syntax: JLPAR [Parameter list]

The system parameters occur only once in the Script.

The parameter list contains the following parameters in this order:

	Values margin	Unit	
1	Print Height Y	0 ... 100	
2	PrintGo-signal	0 - 2	
3	PrintGo-Gate	0 - 2	
4	Encoder-signal	0 - 3	
5	Speed	1 – 1000	dm/min, (for int. Encoder-Signal)
6	Rotation	0 90 180 270	Degree, anticlockwise
7	Mirror-inverted	0 1	
8	Encoder resolution	1 - 2000000000	pulse / m
9	Shifting Change of date	00–23:0-59	hours:minutes
10	Backward Lock	0 1	
11	Speed dependant output	0 ... 1000	dm/min

Table 2: Declaration of the JobList-parameter

Print Height

This parameter adjusts the high-voltage at the deflection plates.

100 corresponds to the max. possible print height of the selected printing mode.

PrintGo Signal

PrintGo-Signal	
Internal	= 0
External Lo-Hi-edge	= 1
External Hi-Lo-edge	= 2

Table 3: PrintGo-Signal

PrintGo-Gate Signal

PrintGo-Gate	
Function deactivated	= 0
Hi-Active	= 1
Lo-Active	= 2

Table 4: PrintGo-Gate

PrintGo-Gate is an external input. With that you can block the external and internal PrintGo-signals.

Hi-Active:

The PrintGo-signals can only pass if the input is on **Hi**.

Lo-Active:

The PrintGo-signals can only pass if the input is on **Lo**.

Encoder signals

Encoder-Signal	
Internal	= 0
External, direction 1 (clockwise)	= 1
External, direction 2 (anticlockwise)	= 2
External, both directions	= 3

Table 5: Encoder-Signal

Rotation, mirror-inverted

The fitting position of the print head is adjusted here.
Mirror-inverted is a reflection by the Y-axis.

Encoder resolution

Is the resolution of the connected encoder in pulses/m.

Shifting Change of date

Usually the date is changed at midnight. With this parameter the time can be shifted by up to 23 h and 59 min. The hour- and minute specifications are separated by a colon.

Backward Lock

Is only reasonable if the Encoder signal is on external, direction 1 or direction 2.
Guarantees that the print will be continued at the correct position after a short operation against the print direction.

Speed dependant output

When the speed (from the encoder) is increasing this value, the output "Speed dependant output" will be switched to high signal.

It will be switched off when the speed is falling under 10% from this value.

From job version V01.06.00.10 the value is specified in decimeter/min so that the resolution will be slightly larger. In the display the value appears in m/min, but with a fractional point.

BEGINJOB parameter list:

Syntax: **BEGINJOB [Parameter list]**

Several jobs can be found in a Script.

Basically a job consists of congeries of objects.

The parameter list of BEGINJOB has the following parameters in the following order:

Values margin	Unit
ID-number	0 - 1023
Name	(alphanumeric text)

Table 6: Parameter list of BEGINJOB

ID-number:

The identification-number can be released.

For double numbers only the last number is considered.

ENDJOB parameter list

The parameter list of ENDJOB is empty.

JOBPAR parameter list (Job parameter)

Syntax: **JOBPAR** [Parameter list]

Values margin		Unit	
1	PrintGo-delay	0 - MAXINT	µm
2	PG-repetition	0 - 65535	
3	PG-distance	0 - MAXINT	µm, only for PG-repetition. > 0:
4	Stroke distance	0 – MAXINT	µm
5	Print Mode	0-10	
	optional:		
6	Flight time compensation	0-MAXINT	
7	Sensor monitoring	0 1	Leave that 0 !
8	Stroke monitoring	0 1	
9	PrintGo monitoring	0 1	
10	PGHoldOff	0-MAXINT	
11	IdentNr CameraJob	(-1) – 2 ³¹	
12	CameraJob Name	(alphanumeric text)	
13	Max. products between LJ3 and Camera	1 - 1000	
14	Product offset between LJ3 and Camera		
15	Cameratrigger-delay	0-MAXINT	µm
16	Print stop interrupts print-out immediately	0 1	
17	Indexcounter of previous camera job data	0 - 65535	
18	Plotmode	0 1	

Table 7: Declaration of the job-parameters

PrintGo-delay

The print (job) starts only after this delay.
 The unit is µm.

PG-repetition

n = 0:	No repetition, that means it is printed once per PrintGo-signal.
1 <= n <= 65534:	Number of prints= n+1
n = 65535:	Number of prints= endless

PG-distance

If PG repetition>1 has been selected, the distance between two prints is indicated in µm with this parameter.

Stroke distance

Distance between two strokes in µm. This value assigns basically the font width.

Print Mode

- 0 = 32 Dot High Quality
- 1 = 32 Dot Fast
- 2 = 32 Dot High Speed
- 3 = 24 Dot High Quality
- 4 = 24 Dot Fast
- 5 = 24 Dot High Speed
- 6 = 24 Dot High Speed Barcode
- 7 = 16 Dot Fast
- 8 = 16 Dot Speed
- 9 = 12 Dot High Speed
- 10 = 5 Dot High Speed
- 11 = 2 Lines High Speed

Until V1.6.0.3

- 0 = 32 Dot High Quality
- 1 = 32 Dot Fast
- 2 = 32 Dot High Speed
- 3 = 24 Dot High Quality
- 4 = 24 Dot Fast
- 5 = 24 Dot High Speed
- 6 = 24 Dot High Speed Barcode
- 7 = 16 Dot Fast
- 8 = 12 Dot High Speed
- 9 = 5 Dot High Speed
- 10 = 2 Lines High Speed

Until V1.04:

- 0 = 32 dot high quality
- 1 = 24 dot high quality
- 2 = 16 dot high speed
- 3 = 7 dot high speed
- 4 = 5 dot high speed
- 5 = 2 Lines High Speed

Flight time compensation

Turned off = 0
 Turned on > 0

Is this value higher than zero, the PrintGo-delay is reduced depending on the speed. Due to this you can obtain that the print is always at the same place even at different speeds. If the value is 0, then the function is turned off.

Sensor Monitoring

Turned off = 0
 Turned on = 1

With this switch 3 monitoring (controls) will be activated:

1. PrintGo Sensor

For this monitoring a 2. PrintGo Sensor will be used. This must be connected to the PrintGo Gate input. Both sensors must be PNP sensors.

The sensors must be adjusted in this way, that the signals of both sensors overlap for a short time.

An error will be generated, when one of the 2 sensors has not switched, when the other has switched on and off.

Monitoring 2 and 3:

An error will be generated too, when there is a PrintGo signal and

- the Production speed is 0 (no encoder pulses, monitoring 2) or
- the print is not started (monitoring 3).

Stroke monitoring

Turned off	= 0
Turned on	= 1

If the stroke frequency is higher than the Leibinger-Jet possible maximum stroke frequency, an error is caused. The print will not be stopped.

PrintGo monitoring

Turned off	= 0
Turned on	= 1

If a further PrintGo-signal appears during the delay or during the print, an error is caused. The print will be not stopped.

PGHoldOff

Value to get a minimum distance of print to print (e.g. for additional printmarks) that means if additional PrintGo-signals appear within the holdoff distance, they will be ignored.

IdentNr CameraJob

-1 = means, that no camera will be used.

0 = means, that the Ljet3 job has no allocation to a configured job on the camera

The Ids (1 – 2³¹) will be generated on the camera host and are downloaded from the Ljet3.

CameraJob Name

The name will be generated on the camera host and downloaded from the LJet3.

Max. products between LJ3 and camera

Control value (Trigger control): Maximum number of products, which are allowed between print head and the camera.

Product offset between LJ3 and camera

Number of products as a fix value, which are between the current printed product and the camera.

Cameratrigger-delay

The trigger signal for the connected camera will be delayed with this value.
The unit is μm .

Print stop interrupts print-out immediately

Can have the value 0 or 1:

- 0: The print stop will be done after the Text-End. The result is that the Text will not be truncated.
- 1: The print stop will be done immediately

Indexcounter of previous camera job data

If no Camera using, the value has to be 0.

The index is always counted by updating a camera job. This is an information for the JETvisio program on the PC. The Index will be compare with the actual loaded job on the JETvisio program. If it is higher, the job will be also updated on the PC.

Plotmode

In the Plotmode the Leibinger-Jet prints always only the first stroke (Stroke 1).
The Plotmode will be stopped with a Printstop or a change of the Job.

Attention: In the Plotmode the Leibinger-Jet is not able to make a phasing.

CNT parameter list (Counter parameter)

Syntax: CNT [Parameter list]

Every counter which is declared refers to the last defined object. The counter is placed at the position where you can find the object {c} in the text.

The parameter list contains the following parameters in this order:

		Values margin
1	Number of digits	1 - 10
2	Start value	0 – 9.999.999.999
3	Initial value	0 – 9.999.999.999
4	End value	0 – 9.999.999.999
5	Increment	-100 ... +100
6	Repetitions	1 - 101
7	Leading zeros	0 1
8	Counting event	0 - 2
9	Counter basis	10 (currently only 10 permitted)
10	Reset	0 - 2
11	LoopMode	0 1 2
12	Repetitor	0 - 100
13	Don't print	0 1
14	Global counter	0 1

Table 8: Declaration of counter

Start value:

This value the counter uses when he reached the end value.

Initialisation value:

This value the counter uses after a jobload or a counter reset.

End value:

When the counter reached this value, he begins with the start value.

Increment:

This value will be added to the counter value when the count event happened.

Repetitions:

Repetition of the actual counter value.

With a repetition of 3 the result can be like this: 1 1 1 2 2 2 3 3 3 ...

Counting event:

After PrintGo = 0
 After PrintActive = 1

With ext. counting input = 2

After PrintGo:

It is counted after every PrintGo signal irrespectively of if a print has been generated or not (similar to MeterGo - funktion at LJ2)

After PrintActive:

It is counted further after every started print (standard).

With ext. Counting input

It is counted further with every rising edge at the X4.2 input.

Reset

Manual	= 0
PC	= 1
External signal	= 2

Manually:

Counter is reset by manual operation or by command from external interface.

PC

Counter is reset synchronously to the product counter, that means if it is reset to 0 by the operator, also the object counter is reset.

External signal

Counter is reset to X4.1 after rising edge.

LoopMode

Print Stop at the end	= 0
Loop	= 1
Up-/Down	= 2 (Not supported by GUI)

Repetitor

Actual value of the done repetitions.

Don't print

When the counter is used for a control function, it can be, that he should not be printed. Then this value is set to 1.

Global counter

When this value is set to 1, this counter is a global counter. In the Jet3 exists one global counter. Its value can be accessed over this button.

COD Parameter list (Barcode parameter)

Syntax: COD [Parameter list]

The parameter list contains the following parameters in this order:

	Values margin	Unit	
1	Barcode type	1 - 10	
2	Bar width 1	1 – 10	Stroke
3	Bar width 2	1 – 10	Stroke
4	Bar width 3	1 – 10	Stroke
5	Bar width 4	1 – 10	Stroke
6	Gap width 1	1 – 10	Stroke
7	Gap width 2	1 – 10	Stroke
8	Gap width 3	1 – 10	Stroke
9	Gap width 4	1 – 10	Stroke
10	Character gap (ICG)	1 – 10	Stroke
11	Barcode height	1 – 32	Ink dots
12	Font for clear figure	(Font name)	
13	Position of clear figure	0 – 4	
14	Calc Check digit	0 - 1	Not used
15	Invers	0 . 1	

Table 9: Declaration of barcode

Barcode type:

Code39	= 1
Code2/5	= 2
EAN8	= 3
EAN13	= 4
UPC A12	= 5
UPC E 8	= 6
128b	= 7
128c	= 8
Postnet	= 9
Datamatrix ECC200 square	= 10
EAN128	= 11
EAN/ECC200 Datamatrix square	= 12
Datamatrix ECC200 rectangular	= 13
EAN/ECC200 Datamatrix rectangular	= 14
USPS 4CB	= 15
QR Code	= 16
PPN Code square	= 17
PPN Code rectangular	= 18

Note:

At the ECC200 the dot- and gap size can be set to 1, 2 or 3. To do this use the parameter Bar width 1.

This control character can contain the types 11, 12 and 14:

{f} = FNC1. This will be replaced in the code by the byte HexE8(Dez232).

This control characters can contain the types 17 and 18:

{g} = Field Separator Gs. Will be replaced in the code by the byte Dez29

The header and the trailer will be added automatically by the Jet3.

Position of clear figure

No	= 0
below	= 1
behind	= 2
in front	= 3
above	= 4

Invers:

The barcode will be printed invers and the clear figure not.

Invers OBJ	Invers COD	Meaning
0	0	Normal print
0	1	Clear figure normal, BC invers
1	0	Clear figure invers, BC invers
1	1	Clear figure invers, BC normal

RPLFIG Parameter list

Syntax: RPLFIG [Parameter list]

The figures of the object are replaced by another character. These replacements refer to the counter fields and the time fields hour, minute, date and calendar week.

The parameter list contains the following substitute characters in this order:

Values margin		Unit
Text field with the following character string		
Substitute character for 0	All available characters	ASCII-Code as Dec.number
Substitute character for 1.	do.	
Substitute character for 2	do.	
Substitute character for 3	do.	
Substitute character for 4	do.	
Substitute character for 5	do.	
Substitute character for 6	do.	
Substitute character for 7	do.	
Substitute character for 8	do.	
Substitute character for 9	do.	
Substitute character for A	do.	//due to reasons of compatibility for
Substitute character for B	do.	// possibly (Hexadecimal counter)
Substitute character for C	do.	
Substitute character for D	do.	
Substitute character for E	do.	
Substitute character for F	do.	

Table 10: Declaration of replacements

RPLDAY Parameter list

Syntax: **RPLDAY [Parameter list]**

The weekdays of the date object are replaced by any printing text.
 The parameter list contains the substitute texts in this order:

Values margin	Unit	
Substitute text for Monday	All available characters	Text
Substitute text for Tuesday	All available characters	Text
Substitute text for Wednesday	All available characters	Text
Substitute text for Thursday	All available characters	Text
Substitute text for Friday	All available characters	Text
Substitute text for Saturday	All available characters	Text
Substitute text for Sunday	All available characters	Text

Table 11: Substitute texts for weekdays

RPLMON Parameter list

Syntax: **RPLMON [Parameter list]**

The months of a date object are replaced by any printing text.
 The parameter list contains the substitute texts in this order:

Values margin	Unit	
Substitute text for January	All available characters	Text
Substitute text for February	All available characters	Text
Substitute text for March	All available characaters	Text
Substitute text for April	All available characters	Text
Substitute text for May	All available characters	Text
Substitute text for June	All available characters	Text
Substitute text for July	All available characters	Text
Substitute text for August	All available characters	Text
Substitute text for September	All available characters	Text
Substitute text for October	All available characters	Text
Substitute text for November	All available characters	Text
Substitute text for December	All available characters	Text

Table 12: Substitute texts for months

RPLMDAY Parameter list

Syntax: RPLMDAY [Parameter list] (as of scriptversion V01.06.00.12)

The days (1...31) of a date object are replaced by any printing text.
 The parameter list contains the substitute texts in this order:

Wertebereich	Einheit	
Substitute text for 01	All available characters	Text
Substitute text for 02	All available characters	Text
Substitute text for 03	All available characters	Text
Substitute text for 04	All available characters	Text
Substitute text for 05	All available characters	Text
.....		
Substitute text for 31	All available characters	Text

Table 12.1: Substitute texts for days

Example script with replacements from months and days:

```

BEGINLJSCRIPT [(V01.06.00.12)]
JLPAR [90 0 0 3 60 0 0 10000 00:00 0 7000]
BEGINJOB [ 0 () ]
JOBPAR [ 0 0 0 250 0 0 0 1 1 0 -1 () 1 0 55000 1 0 ]
OBJ [1 2 24 0 (ISO1_7X5) ({t}) 1 0 0 0 0 1 0 0 0 0 0 0 () () 0 0 () ]
TIME [ (dd.mm.yyyy) 0 1 0 ]
RPLDAY [ (MON) (TUE) (WED) (THU) (FRI) (SAT) (SUN) ]
RPLMON [ (01) (02) (03) (04) (05) (06) (07) (08) (09) (10) (11) (12) ]
RPLYEAR [ 2013 (2013) (2014) (2015) (2016) (2017) (2018) (2019) (2020) (2021) (2022) (2023) (2024) (2025) (2026)
(2027) (2028) (2029) (2030) (2031) ]
OBJ [2 2 13 0 (ISO1_7X5) ({t}) 1 0 0 0 0 1 0 0 0 0 0 0 () () 0 0 () ]
TIME [ (dd.mm.yyyy) 0 1 0 ]
RPLDAY [ (MON) (TUE) (WED) (THU) (FRI) (SAT) (SUN) ]
RPLMON [ (01) (02) (03) (replace_month) (05) (06) (07) (08) (09) (10) (11) (12) ]
RPLYEAR [ 2013 (2013) (2014) (2015) (2016) (2017) (2018) (2019) (2020) (2021) (2022) (2023) (2024) (2025) (2026)
(2027) (2028) (2029) (2030) (2031) ]
RPLMDAY [ (01) (02) (03) (04) (05) (06) (07) (08) (09) (10) (11) (12) (13) (14) (15) (16) (17) (18) (19) (20) (21) (22)
(23) (24) (25) (26) (27) (28) (29) (30) (31) ]
OBJ [3 2 2 0 (ISO1_7X5) ({t}) 1 0 0 0 0 1 0 0 0 0 0 0 () () 0 0 () ]
TIME [ (dd.mm.yyyy) 0 1 0 ]
RPLDAY [ (MON) (TUE) (WED) (THU) (FRI) (SAT) (SUN) ]
RPLMON [ (01) (02) (03) (04) (05) (06) (07) (08) (09) (10) (11) (12) ]
RPLYEAR [ 2013 (2013) (2014) (2015) (2016) (2017) (2018) (2019) (2020) (2021) (2022) (2023) (2024) (2025) (2026)
(2027) (2028) (2029) (2030) (2031) ]
RPLMDAY [ (01) (02) (03) (04) (05) (06) (07) (08) (09) (replace_day) (11) (12) (13) (14) (15) (16) (17) (18) (19) (20)
(21) (22) (23) (24) (25) (26) (27) (28) (29) (30) (31) ]
ENDJOB []
ENDLJSCRIPT []
  
```

Result of print:

```

10.04.2014
10.replace_month.2014
replace_day.04.2014
  
```

RPLYEAR Parameter list

Syntax: RPLYEAR [Parameter list]

The years of a date object are replaced by any printing text.
 The parameter list contains the substitute texts in this order:

Values margin	Unit	
Basis year	(e.g.) 2000	Year
Substitute text for basis year	All available characters	Text
Substitute text for basis year+1	All available characters	Text
Substitute text for basis year+2	All available characters	Text
Substitute text for basis year+3	All available characters	Text
...	...	...
Substitute text for basis year+18	All available characters	Text

Table 13: Substitute texts for years

SHIFTS Parameter list *until LJScriptversion V1.6.0.0*

Syntax: SHIFTS [Parameter list]

5 shifts can be administrated. The starting time and the printing text are defined here for every of these 5 shifts.

The parameter list contains the following parameters in this order:

Name of the Parameter	Content	Unit
Starting time shift 1	00:00 - 23:59	hh:mm
Starting time shift 2	00:00 - 23:59	hh:mm
Starting time shift 3	00:00 - 23:59	hh:mm
Starting time shift 4	00:00 - 23:59	hh:mm
Starting time shift 5	00:00 - 23:59	hh:mm
Printing text for shift 1	(Text)	
Printing text for shift 2	(Text)	
Printing text for shift 3	(Text)	
Printing text for shift 4	(Text)	
Printing text for shift 5	(Text)	

Table 14: Declaration of shifts

SHIFTS Parameterliste *from LJScriptversion V1.6.0.1*

Syntax: SHIFTS [Parameterliste]

Until 24 shifts can be administrated. The starting time and the printing text are defined here for every of these shifts.

The parameter list contains the following parameters in this order:

Name of the Parameter	Content	Unit
Number of shifts	1 ... 24	
Starting time shift 1	00:00 - 23:59	hh:mm
Starting time shift 2	00:00 - 23:59	hh:mm
Starting time shift 3	00:00 - 23:59	hh:mm
...	00:00 - 23:59	hh:mm
Starting time shift n	00:00 - 23:59	hh:mm
Printing text for shift 1	(Text)	
Printing text for shift 2	(Text)	
Printing text for shift 3	(Text)	
...	(Text)	
Printing text for shift n	(Text)	

Table 14: Declaration of shifts

1 <= n <= 24

EXTTXT Parameter list

Syntax: EXTTXT[Parameter list]

The before defined object is defined as ExternText. The external text is then fitted at the place where {e} is inserted in the text string.

The parameter list contains the following parameters in this order:

Values margin	Unit
1 Length	1-100
2 Placeholder	(Text characters)
3 Field number 0=ExtText, 1...x Field number Mailmode	
4 In ExtText: Character offset In Mail-Mode: Not used	0-255
5 Check double prints In Mail-Mode: Not used	0, 1

Table 15: Declaration of extern text parameter list

By the field number you can decide between an Extern-Text-Variable and a „Mail“-Variable. 0=ExtText-Variable, 1...x=Mail-Variable, the value corresponds then to the field number.

Extern-Text-Variable:

The number of placeholders defines the number of external characters for this object.

The character offset defines from which character of the external text this object will be filled with characters.

The LJ3 calculates independantly the required characters for the actual loaded job. After he received completely the last character all will be inserted automatically into to print text. To calculate this there will be built the sum for each object from the character offset and the placeholders. The largest sum is the value for required characters of the loaded job.

In the Extern Text Mode the function “Check double prints” can be used. When after a print go signal there is not one or more received external text, the print will be stopped and an error message is generated.

TIME Parameter list

Syntax: **TIME [Parameter list]**

The time and date functions which are referenced in the text string of the object are formatted by this declaration. The date is then fitted at the place where {t} is inserted in the text string.

The parameter list contains the following parameters in this order:

Values margin		Unit
1	Format	(Text)
2	Expiry offset days	Days (0 – 30000)
3	Leading zeros	0 1
4	Expiry offset months	Months (0 – 1000)
5	Special replacements for hours and minutes	0 1

Table 16: Declaration of time-variables

Layout Format

d	Day	X
m	Month	
y	Year	4 digits
Y	Year	X
H	Hour	X
M	Minute	X
S	Second	X
w	Weekday	
j	Julian date	X
c	Calender week	X
f	F2897	

If this variable is listed several times the complete passage will be replaced by the substitution of the value.

For the variables which are indicated with „X“ the inserted value is always displayed double-digit (if applicable with leading „0“), if the variable is set twice.

Starting from Scriptversion V1.6.0.2 the leading zeros will be set by parameter 3.

0 = without leading zeros

1 = with leading zeros

For the other variables the length is defined by the replacements.

F2897:

The julian date and the 2 digits year are concatenated and from this up to 5 digits decimal value will be calculated a 3 digits value in the base 62 (with left-hand zeros0).

Starting from Scriptversion V1.6.0.9

it is possible to choose the expiry offset between days or months.

0 = disabled (no offset)

1 – maximum value = enabled

If parameter 2 (expiry offset days) > 0, it is important to set parameter 4 (expiry offset month) to zero.

Otherwise, if parameter 4 (expiry offset months) > 0, it is important to set parameter 2 (expiry offset days) to zero.

Starting from Scriptversion V01.06.00.13

With parameter 5 it is possible to choose special replacements for hours and minutes.

0 = disabled (no special replacements)

1 = hours and minutes will be replaced like following table:

hours	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z

Minutes	0 - 9	10 - 19	20 – 29	30 - 39	40 - 49	50 - 59
	a	b	c	d	e	f

Example script with special replacements:

```
BEGINLJSCRIPT [(V01.06.00.14)]
JLPAR [90 1 0 3 30 0 0 50000 00:00 0 7000]
BEGINJOB [ 0 () ]
JOBPAR [ 0 0 0 300 0 0 0 1 1 0 -1 () 1 0 55000 1 0 ]
OBJ [1 1 0 0 (ISO1_7X5) ({t}) 1 0 0 0 0 1 0 0 0 0 0 () () 0 0 () ]
TIME [ (HH:MM) 0 1 0 0 ]
RPLDAY [ (MON) (TUE) (WED) (THU) (FRI) (SAT) (SUN) ]
RPLMON [ (01) (02) (03) (04) (05) (06) (07) (08) (09) (10) (11) (12) ]
RPLYEAR [ 2014 (2014) (2015) (2016) (2017) (2018) (2019) (2020) (2021) (2022) (2023) (2024) (2025) (2026) (2027)
(2028) (2029) (2030) (2031) (2032) ]
OBJ [2 1 14 0 (ISO1_7X5) ({t}) 1 0 0 0 0 1 0 0 0 0 0 () () 0 0 () ]
TIME [ (HH:MM) 0 1 0 1 ]
RPLDAY [ (MON) (TUE) (WED) (THU) (FRI) (SAT) (SUN) ]
RPLMON [ (01) (02) (03) (04) (05) (06) (07) (08) (09) (10) (11) (12) ]
RPLYEAR [ 2014 (2014) (2015) (2016) (2017) (2018) (2019) (2020) (2021) (2022) (2023) (2024) (2025) (2026) (2027)
(2028) (2029) (2030) (2031) (2032) ]
ENDJOB []
ENDLJSCRIPT []
```

Result of print:



OBJ Parameter list (Object parameter)

Syntax: OBJ [Parameter list]

An object is the smallest unit in the description of a print.

The parameter list contains the following parameters in this order:

Values margin		Unit	
1	ID-number	1 - 32	
2	Position x	0 - 19999	Strokes
3	Position y	0 - 31	Ink dots
4	LType	0, 1	Language type
5	Font	(Font name)	
6	Printing text	(Text)	ASCII / Unicode, determination by Font
	optionally:		
7	Character gap	0 – 7	Strokes
8	Reserve	0	Leave it 0
9	Reserve	0	Leave it 0
10	Rotation	0 90 180 270	Degree, anticlockwise
11	Mirror-inverted	0 1	
12	Proportional	0 1	
13	Bold text	0 - 7	
14	Contrast print	0 - 7	
15	Inverse	0 1	
16	Alternating	0 1	
17	Camera field	0 –2	Only two camerafields per job, otherwise zero
18	Camera teach mode	0 –255	Only in text- and extern text objects, otherwise zero
19	Camera teach text	(Text)	Only in text- and extern text objects, otherwise empty string
20	Keyboardname	(Text)	Language of your Keyboard
21	Counter-dependent	0 - 32	Counter ID-No.
22	Prompt-Field max. sign	max. Length of each object	Prompt-Field Max character from the edited field
23	Prompt-Field name	(Text) max. 40 sign	Users text for each Prompt-Field

Table 17: Declaration of objects

ID-number:

The identification-numbers can be assigned in any order.

For double numbers only the last one is considered.

Position:

The position of an object refers to the position of the job which calls this object. If the object is linked, the position x refers to the end of the previous object.

Ltype:

For language specific handling of textes

0 = Standard

1 = Arabic

Fonts:

Every font is only identified by its name. Unicode fonts have the sign ~ before the font name. The LJ3 has a number of fixed available fonts.

Fonts can be downloaded or also deleted.

All fonts are Bitmap Fonts.

If the character string (*\$CODE*) is entered for the font, the object is started as barcode object which has to be defined by a further element COD [...].

If the character string (*\$GRAFIC*) is entered, the object is started as graphic object and the graphic in the printint text has to be defined as described below.

Printing text for the function graphic

The printing text has 3 parameters.

(Graphic height Graphic width Graphic) Separation character is a blank

Values margin	Unit	
Graphic height	1 – 32	Ink dots
Graphic width	1...20000	Ink dots (Strokes)
Graphic	0...9 A...F	Bit coded as hex number in ASCII

Table 18: Printing text for the function graphic

Layout of the bit coded graphic:

One bit is used per pixel so that 8 bits can be saved in one byte.

Up to 4 Hex-Bytes or 8 ASCII-Bytes can be needed for one stroke. The actual amount of bytes per stroke depends on the graphic height. If the graphic is e.g. 17 dots high, 3 Hex-Bytes are used per stroke. .

Graphic:

Byte 1 Stroke 1

Byte 2

...

Byte n

Byte n+1 Stroke 2

Byte n+2

...

Byte 2*n

Byte 2*n+1 Stroke 3

Byte 2*n+2

...

n = Amount of bytes per stroke

Pixel Y.1 reserves Bit 0 of the first Hexbyte.

Pixel Y.2 reserves Bit 1 of the first Hexbyte.

...

Pixel Y.8 reserves Bit 7 of the first Hexbyte.

Pixel Y.9 reserves Bit 0 of the second Hexbyte.

...

As the graphic is transferred in ASCII-characters, two ASCII-Bytes are used for one Hex-Byte.

Example for graphic:

(12 4 FF0F01080108FF0F)

This graphic is 12 dots high, that means $n = 2$

X.1 (Stroke 1): All points Y.1..12 are set

X.2 (Stroke 2): Point Y.1 and point Y.12 are set

X.3 (Stroke 3): Point Y.1 and point Y.12 are set

X.4 (Stroke 4): All points Y.1..12 are set

This graphic shows a rectangle.

Printing text

The text which is shown in the parenthesis is printed. The parenthesis will be not printed.

Character gap

Shows the distance between the several characters. Default value is 1.

Rotation, Mirror-inverted

Every object can be rotated/mirrored.

Mirror-inverted is a reflection by the Y-axis.

Proportional:

Every font can be printed proportional or not proportional. In principle the fonts are applied proportional.

For non-proportional print the RIP (printer) inserts empty strokes up to the maximum character width. They will be inserted that the character is placed centered.

User-specific graphics are stored in one or more special fonts.

Bold text:

Every stroke is printed as often as given by the indicated number, but every stroke is printed separately. The character width increases accordingly. The bold text has no influence on the maximum printing speed.

Contrast print:

Every stroke is printed as often as given by the indicated number, but the additional strokes are printed on top of the previous stroke. The character width does not increase, but the maximum printing speed reduces.

This parameter will be taken over automatically to all objects, which are located in this X-area. The highest contrast print is carried out.

Inverse:

All pixels are inverted.

An ink dot will not become an ink gap.

an ink gap dot will become an ink dot.

If required an additional frame is drawn around the object (e.g. for texts, special barcode types)

Alternate:

After every PrintGo the print it is turned by 180°.

Prompt-Field:

The prompt-field function can only use in following objects (with limitations):

- Text-Object (text content is editable)
- Barcode-Object (text content is editable)
- Counter-Object (counter end value is editable)
- Date-Object (only editable, if declared as expiration date)

Construct:

Parameter 22 describes the maximum length of the editable content

Parameter 23 describes the name in plain writing of the prompt field.

PGJOB Parameter list (Batchjob)

Syntax: PGJOB [Parameter list]

Every PrintGo is assigned to a job.

The parameter list contains the following parameters in this order:

Values margin	Unit	
ID-No.	1 - 99	
Number Repeats	0 – 99.999.999	0 = print only once
Job-ID-No.	Available jobs	

Table 19: PGJOb parameter list

If you are working with PGJOB, the call of EXTSEL and JOBORg (see below) is not permitted.

ID-number:

These numbers are allocated in ascending order to process them consecutively. The number has to start with 1 and has to be continuously without any gap.

Number of repeats:

Number of repetitions of the jobs which are indicated with the job-ID-number.

For 0 it is printed only once (no repetitions).

Job-ID-No:

ID No of the assigned and available jobs (see ID-number for BEGINJOB).

EXTSEL Parameter list (Extern Job Select)

Syntax: EXTSEL [Parameter list]

Via digital inputs you can select a job which is printed with the next Print Go.

If you work with EXTSEL the call with PGJBO and JOBORg is not permitted.

The parameter list contains the following parameters in this order:

Values margin	
External applied number	0 - 1023 x X
Job-ID-No	Available Job-No.

Table 20: Declaration of the external inputs for job selection

External applied number:

With 10 digital input signals you can apply the numbers 0 up to 1023.

Every number which can occur should be assigned to a job with this command. The job which is assigned to the external number X is printed. If numbers are applied for which no job is declared the previous Job will be printed again.

Job-ID-No:

ID No of the assigned and available jobs (see ID-number for BEGINJOB)

JOBORG Parameter list (Job organizer)

Syntax: **JOBORG [Parameter list]**

After a PrintGo the listed jobs are printed in the given distances.
 The parameter list contains the following parameters in this order:

Values margin	Unit	
ID-No	1 - 99	
Distance	0 - 999999	µm, 0=directly after previous job
Job-ID-No.	Available jobs	

Table 21: Allocation of job-organizer

If you work with JOBORg the call with EXTSEL and PGJOB (see above) is not permitted.

ID-number:

These numbers are allocated in ascending order to process them consecutively. The number has to start with 1 and has to be continuously without any gap.

Distance:

The job is printed in the given distance after the PrintGo. If 0 is given, the job is printed immediately after the previous job.

Job-ID-No:

ID No of the assigned and available jobs (see DI-number for BEGINJOB)

If the joblist has to be repeated without extern PG-Signal, so the distance between jobstart1 and jobstart2 and so on must be set in the PG-Delay of the first job. Even the count of repeats has to be set in the PG-repetition of the first job.

EXAMPLE for a Script:

```

BEGINLJSCRIPT [(V1.1.0.0)]
JLPAR [85 0 0 0 2 0 0 25000 00:00 0]
BEGINJOB [ 0 ( ) ]
JOBPAR [ 20000 0 0 350 0 0 0 1 1 0]
OBJ [0 1 0 0 (ISO7_24x18) (Text) 1 0 0 0 0 1 0 0 0 0]
OBJ [0 62 0 0 (ISO7_12X8) ({c}) 1 0 0 0 0 0 0 0 0 0]
CNT [ 4 1000 0 8000 1 1 1 1 10 0 1]
OBJ [0 99 0 0 (ISO7_7X5) ({t}) 1 0 0 0 0 0 0 0 0 0]
TIME [ (dd.mm.yyyy) 0]
RPLDAY [ (MON) (TUE) (WED) (THU) (FRI) (SAT) (SUN) ]
RPLMON [ (JAN) (FEB) (MAR) (APR) (MAY) (JUN) (JUL) (AUG) (SEP) (OCT)
(NOY) (DEC) ]
ENDJOB []
ENDLJSCRIPT []

```

Resulting imprint:

Text000020.MAY.2008

Object 1: Fix text

Object 2:
CounterObject 3: Date
(Day:Month:Year)

History

V1.01, Page 20

Ltype bei Obj Parameterliste hinzugefügt

V1.02, Page 17

Replacements for years (RPLYEAR) increased to 19

V1.03, Page 18

Extern Text: Parameter character offset added

V1.04, Page 9+13+19, Oct.19.2009

JOBPAR. 11: Cameratrigger-delay added

OBJ. 17: Camera field added

V1.05, Page 10 11.03.2010 from V60.0.01.1

Some PrintModes added

V1.06, Page 9+20 22.03.2010 from V60.0.01.2

Printparameter 16 added

Printparameter 17 added

Obj. parameter ID18 added

Obj. parameter ID19 added

V1.6.0.1, Page 18 April, 22.2010 from V60.0.1.2

Shifftimes increased to 24 and variabel designed.

V1.6.0.2, Page 9 + 11 + 19 03.05.2010 from V60.0.1.3

Obj. parameter ID3 added

Printparameter 16 was wrong

Jobparameter 11 improved

V1.6.0.3, Page 18

Extern Text: Parameter „Check double prints“ added.

V1.6.0.4, Page 10: June, 20.2011 from V60.0.01.08

New printmode: 16 Dot Speed

V1.6.0.5, Page 7: 19.10.2011 from V60.0.1.10

New: Speed dependant output

V1.6.0.6, Page 20: Feb., 15.2012 from V60.0.1.12

F2897 added

V1.6.0.7, Page 15: May 07.2012 from V60.0.1.13

USPS barcode added

QR code added

V1.6.0.8, Page 15: June 27.2012 from V60.0.1.14

PPN Code ECC200 square and rectangular added

V1.6.0.9 Oct. 17.2012 from V60.0.1.15

Page 20: Expiry offset in months added

Page 19: Value range from character offset at ExternText increased from 100 to 255

V1.6.0.10 Jan.01.2013 from V60.0.1.17 was shifted to the V60.0.1.18

JLPAR 5 and 11: unit speed changed from m/min to dm/min.

V1.6.0.11 13.02.2013 ab V60.0.1.17 was shifted to the V60.0.1.18

Extension for PromptField

V1.06.0.12 10.04.2013

Bugfix of Description on page 3 and repeats of CNT.

Date/Time object: Replacements for days 1,2,3,4,...,31

V1.06.0.13 18.04.2013 from V60.0.1.19

Description Extension of up-/down mode in LOOP MODUS on counter objects

JOBPAR: Text "Print stop reacts after TextEnd" replaced with: "Print stop interrupts print-out immediately"

Date/time object: It is possible to choose special replacements for hours and minutes.

V01.06.00.14 13.03.2014 from V60.0.24.00

CNT Page 12: new Parameter Global counter

V01.06.00.15 22.04.2014 from V60.0.25.00

JOBPAR: Par. 7 will be used. 17.04.2014 Ab V60.0.25.00

JOBPAR Par. 18 is new